

Programming Exam 4

Date/Time: 2026.04.07 08:00 – 08:50

(程式檔命名: 學號_pe4.py, 上傳至 Moodle PE4 上傳區)

Problem: Reverse Column Zigzag Multiplication Table

Write a Python program that fills a 9×9 grid using the standard 9×9 multiplication table in a **zigzag column-wise pattern from right to left**, and prints each multiplication expression as it is placed.

Rules

- Generate all multiplication expressions from $9*1$ to $1*9$ in the usual order:
 - $9*1, 9*2, \dots, 2*7, 2*8, 2*9, \dots, 1*9$
- Fill the 9×9 grid **column by column**, but start from the **rightmost column** and move toward the **leftmost column**.
- The filling direction alternates by column:
 - The **rightmost column** is filled **top to bottom**
 - The next column is filled **bottom to top**
 - The next column is filled **top to bottom**
 - and so on, until all 9 columns are filled
- Each time an expression is placed into the grid, immediately print it in the format $a*b=c$, padded to width 8 and left-aligned.
- After filling the entire grid, print the final 9×9 grid.

Notes

- There is **no input**.
- Use width 8 for every printed cell in the final grid so that the columns line up neatly.

Sample Output

(以下是你程式執行後須印出的結果)

```
(wq_env) PS C:\Users\user\Desktop\AI Lab\test> python .\test1.py
Reverse Column Zigzag Multiplication Table:
1*1=1  2*9=18 3*1=3  4*9=36 5*1=5  6*9=54 7*1=7  8*9=72 9*1=9
1*2=2  2*8=16 3*2=6  4*8=32 5*2=10 6*8=48 7*2=14 8*8=64 9*2=18
1*3=3  2*7=14 3*3=9  4*7=28 5*3=15 6*7=42 7*3=21 8*7=56 9*3=27
1*4=4  2*6=12 3*4=12 4*6=24 5*4=20 6*6=36 7*4=28 8*6=48 9*4=36
1*5=5  2*5=10 3*5=15 4*5=20 5*5=25 6*5=30 7*5=35 8*5=40 9*5=45
1*6=6  2*4=8  3*6=18 4*4=16 5*6=30 6*4=24 7*6=42 8*4=32 9*6=54
1*7=7  2*3=6  3*7=21 4*3=12 5*7=35 6*3=18 7*7=49 8*3=24 9*7=63
1*8=8  2*2=4  3*8=24 4*2=8  5*8=40 6*2=12 7*8=56 8*2=16 9*8=72
1*9=9  2*1=2  3*9=27 4*1=4  5*9=45 6*1=6  7*9=63 8*1=8  9*9=81
```

(繳交是交 pe4.py, 不是交截圖)

Note: You need to write comments (註解) for each part in your code.